Raphael Karamagi

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Education

Duke University Durham, NC
B.S.E. Electrical & Computer Engineering, B.S. Computer Science, GPA: 3.83 *Aug. 2025 – May 2029*
Selected Coursework: Computer Architecture, Signals & Systems, Data Structures & Algorithms, Probability, Linear Algebra, Multivariable Calculus

Experience

Software Engineering Intern – AI Security & Evaluation May 2026 – July 2026
Duke OIT (Code+ Program) *Durham, NC*

- Building automated security scanning pipeline that audits open-source models from Hugging Face for malicious code and supply-chain vulnerabilities before deployment to Duke systems (60,000+ users).
- Developing evaluation harness with dashboards and risk-scoring analytics to benchmark on-premises vs. cloud-hosted LLMs across task types, guiding enterprise model selection.

Software Engineer Intern July 2024 – Aug. 2024
IntelliSOFT Consulting *Nairobi, Kenya*

- Implemented REST API endpoints with role-based access in OpenMRS EMR; led deployment to 20+ hospitals across South Sudan and 3 Nairobi pilot sites with 98% uptime.

Projects

Stock Prediction Platform v2 | *C++, PyTorch, ONNX, LSTM, FinBERT* Jan. 2026 – Present

- Co-led 5-person team building a dual-model (LSTM + NLP) prediction system across 15 tickers.
- Architecting v2 around a C++ event-driven backtester with order book simulation, transaction cost modeling, and walk-forward validation; PyTorch model exported via ONNX for low-latency inference.

Compost Now Wash Automation | *Arduino, C/C++, MOSFET, Embedded* Jan. 2026 – Apr. 2026

- Led team designing automated shutoff for industrial bin washers, reducing waste & preserving manual operation.
- Designed embedded control circuit (Arduino Nano + 12V solenoid via MOSFETs with flyback protection) and wrote non-blocking C/C++ firmware.

ASL Computer Vision Classifier | *PyTorch, OpenCV, MediaPipe* Oct. 2025 – Dec. 2025

- Built dual-model CV system classifying 29 ASL gestures from 87K+ images (71% test accuracy); designed Landmark NN (∼10K params) for real-time CPU inference with 100% hand-detection accuracy.

Leadership & Activities

Duke Applied Machine Learning – Senior Engineer Sep. 2025 – Present

- Mentor junior members through the AITP program; lead Stock Prediction Platform team.

National Society of Black Engineers (NSBE) – Active Member, Duke Chapter Aug. 2025 – Present

- Represented Duke at Fall 2025 Regional Conference and 2026 Annual Convention (Baltimore, MD).

Technical Skills

Languages: Python, Java, C/C++, R, SQL, JavaScript
ML & Data: PyTorch, TensorFlow, scikit-learn, OpenCV, MediaPipe, pandas, NumPy
Frameworks & Tools: Flask, FastAPI, React, Node.js, Git, Docker, Linux/Bash, Arduino

Honors & Awards

5th Globally, Cambridge AS-Level Economics | UKMT Senior Math Challenge Silver (2024) | SAT 1540/1600